CHAPTER 1:FUNCTIONS AND THEIR GRAPHS

1.1

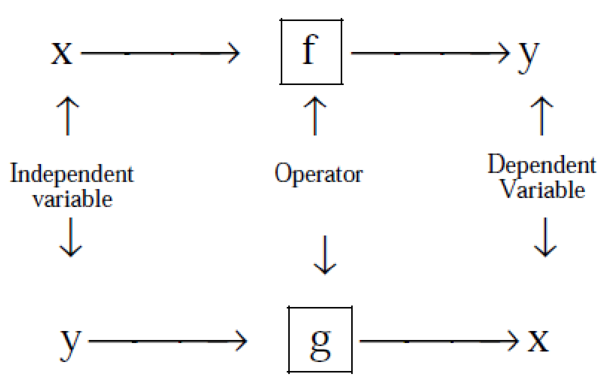
FUNCTIONS

Definition: If to every value (considered as real unless other-wise stated) of a variable x, which belongs to some collection (Set) A, there corresponds one and only one finite value of the quantity **y**, then **y** is said to be a function (Single valued) of **x** or a dependent variable defined on the set A ; **x** is the argument or independent variable.

If to every value of **x** belonging to some set A there corresponds one or several values of the variable **y**, then **y** is called a multiple valued function of **x** defined on A.Conventionally the word **"FUNCTION"** is used only as the meaning of a single valued function, if not otherwise stated .

See a function as an operator, which takes an input and gives an output. The input is called *the independent variable* and the output, which depends on the input because of the operator, is called the *dependent variable*

Pictorially:



Domain, Co-domain & Range Of A Function :

Let $f : A \to B$, then the set A is known as the domain of f & the set B is known as co-domain of f . The set of all f images of elements of A is known as the range of f . Thus :

Domain of $f = \{a \mid a \in A, (a, f(a)) \in f\}$

Range of $f = \{f(a) \mid a \in A, f(a) \in B\}$

It should be noted that range is a subset of co-domain . Sometimes if only f (x) is given then domain is set of those values of ' x ' for which f (x) exists or is defined .

Exercise1.1:

Q1: Given a function $f(x)=\dfrac{x+1}{x-1}$,find $f(2x), 2f(x), f(x^2), (f(x))^2$.

Q2: Given a function $f(x)=\dfrac{4^x}{4^x+2}$,prove that $f(x)+f(1-x)=1$,and also find the value of

$f\left(\dfrac{1}{81}\right)+f\left(\dfrac{2}{81}\right)+f\left(\dfrac{3}{81}\right)..............+f\left(\dfrac{80}{81}\right)$.

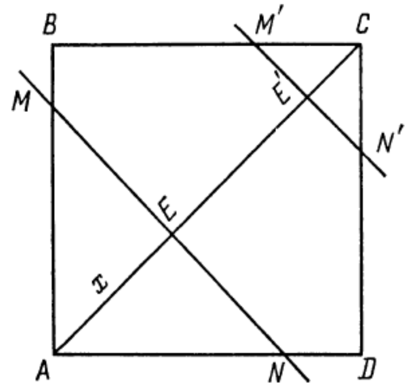
Q3: if $f(x)=\cos(\log x)$, then evaluate $f(x).f(y)-\dfrac{1}{2}\left\{f\left(\dfrac{x}{y}\right)+f(xy)\right\}$

Q4: Given a function $f(x)=\log\dfrac{1-x}{1+x}$ .show that at $x_1, x_2 \in (-1,1)$,the following identity holds true.

$f(x_1)+f(x_2)=f\left(\dfrac{x_1+x_2}{1+x_1x_2}\right)$.Also prove that $f\left(\dfrac{2x}{1+x^2}\right)=2f(x)$.

Q5: Given a function $f(x)=\begin{cases} 3^{-x}-1; -1\le x<0, \\ \tan\left(\dfrac{x}{2}\right); 0\le x<\pi, \\ \dfrac{x}{x^2-2}; \pi\le x<6. \end{cases}$ Find $f(-1), f\left(\dfrac{\pi}{2}\right), f\left(\dfrac{2\pi}{3}\right), f(4)$.

Q6: In the square ABCD with side AB=2 a straight line MN is drawn perpendicular to AC .Denoting the distance from the vertex A to the line MN as x ,express through the x the area S of triangle AMN cut off from line MN .Find the area at x=√2/2 and x=2.



Q7: A function *f* satisfies the condition $f\left(x+\frac{1}{x}\right)=x^2+\frac{1}{x^2}, x \neq 0$. Determine it and hence find *f(3)*

Q8: An polynomial expression *f(x)* of degree n satisfies the condition $f(x)+f\left(\frac{1}{x}\right)=f(x).f\left(\frac{1}{x}\right)$, then prove that $f(x)=1 \pm x^n$

Q9: If for non zero *x*, $a.f(x)+b.f\left(\frac{1}{x}\right)=\frac{1}{x}-5$, where *a≠b*,then *f(x)* is equal to?

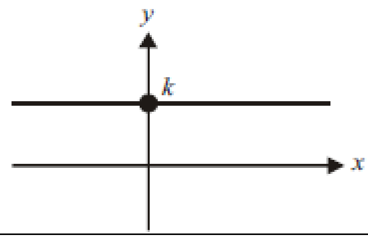
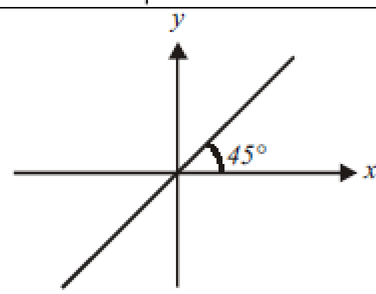
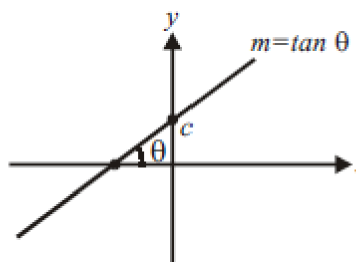
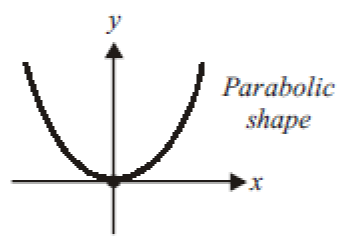
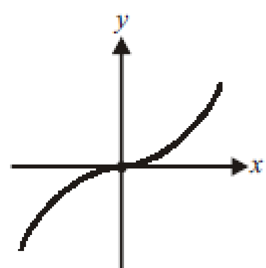
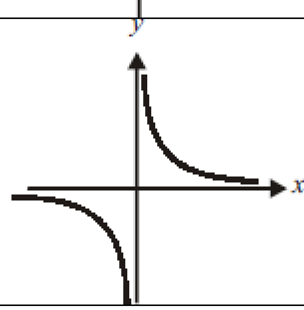
Q10: Let $f_1(x)=\frac{x}{3}+10$, for all x ЄR and $f_n(x)=f_1(f_{n-1}(x))$, for all n≥2.Then find $f_n(x)$.

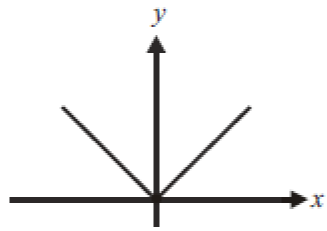
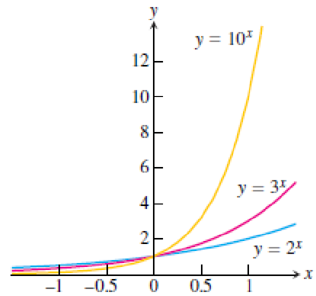
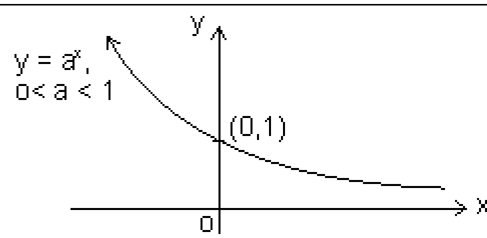
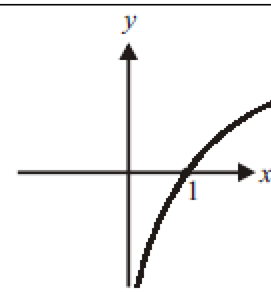
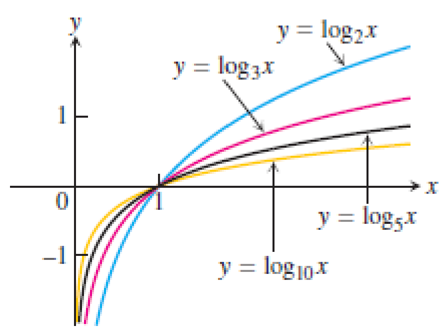
Q11: If $3.f(x)-f\left(\frac{1}{x}\right)=\log_e x^4$,for x>0,Find $f(e^x)$

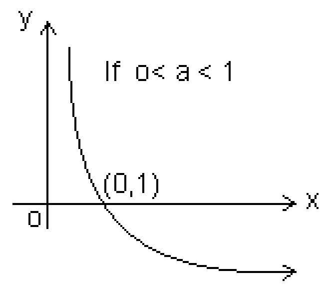
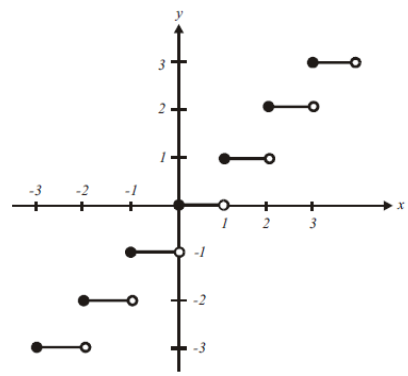
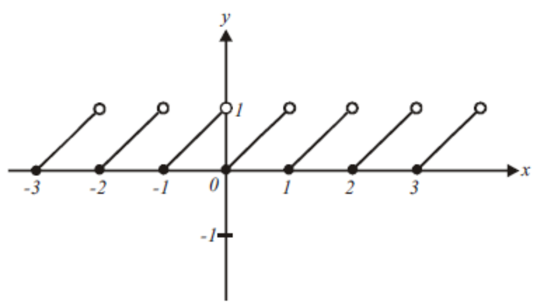
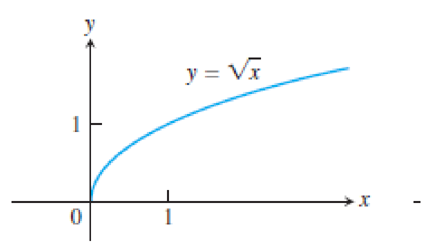
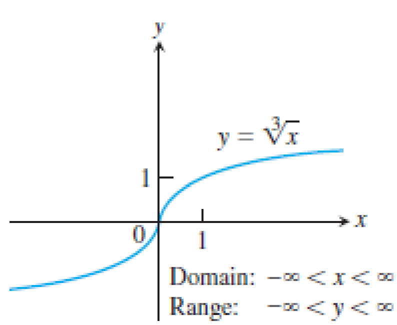
Q12: If $f(x)=\cos\left[\pi^2\right]x+\cos\left[-\pi^2\right]x$ where [x] stands for the greatest integer function, then evaluate f (π / 2), f (π), f (–π) and f (π / 4) .

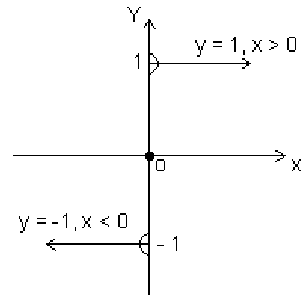
1.2

GRAPHS OF FUCTIONS

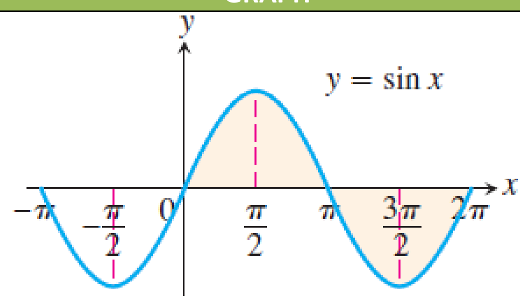
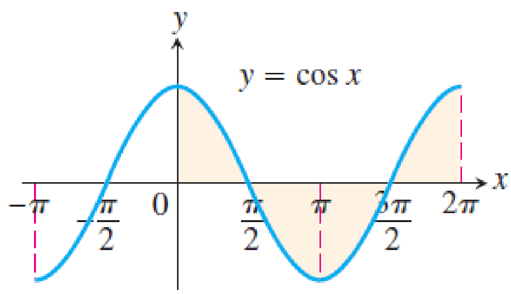
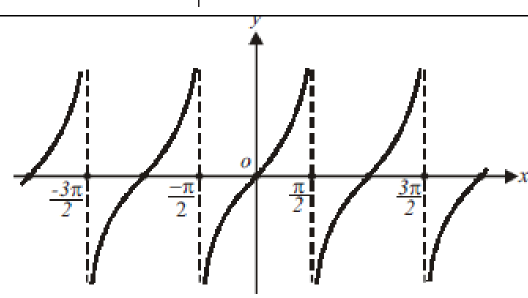
S.NO	FUNCTION		GRAPH	DOMAIN	RANGE
a)	***Constant function***	***f(x)=k***		$\mathbb{R}$	{k}
b)	***Identity function***	***f(x)=x***		$\mathbb{R}$	$\mathbb{R}$
c)	***Linear function***	***f(x)=mx+c***		$\mathbb{R}$	$\mathbb{R}$
d)	***Square function***	***f(x)=x²***		$\mathbb{R}$	$[0,\infty)$
e)	***Cube function***	$f\,(x) = x^3$		$\mathbb{R}$	$\mathbb{R}$
f)	***Reciprocal function***	$f\,(x)=1/x$		$\mathbb{R}-\{0\}$	$\mathbb{R}-\{0\}$

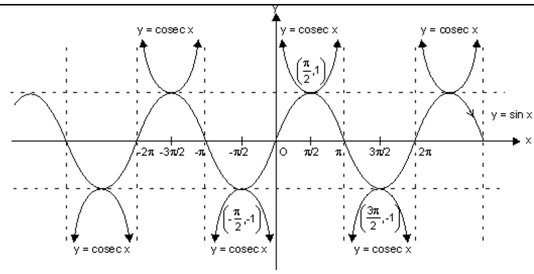
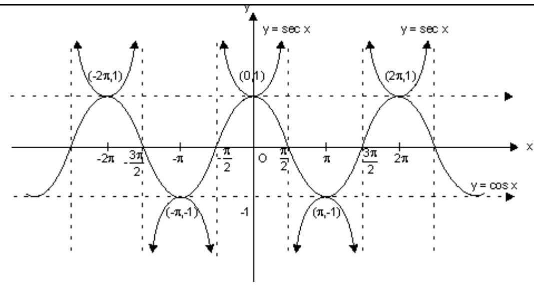
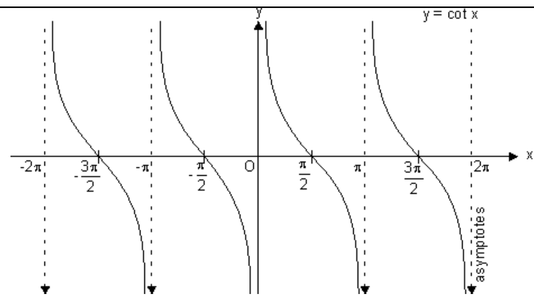
g)	***Modulus Function***	$f(x)=\lvert x\rvert$		$\mathbb{R}$	[0,∞)
h)	***Exponential function***	$f(x)=a^x$ $a>1$		$\mathbb{R}$	(0,∞)
		$f(x)=a^x$ $0<a<1$		$\mathbb{R}$	(0,∞)
i)	***Logarithmic function***	$f(x)=\log_e x$		(0,∞)	$\mathbb{R}$
		$f(x)=\log_a x$ $(a>1)$		(0,∞)	$\mathbb{R}$

		$f(x)=\log_a x$ $(0<a<1)$		$(0,\infty)$	$\mathbb{R}$
j)	***Greatest integer function***	$f(x)=[x]$		$\mathbb{R}$	$\mathbb{Z}$
k)	***Fractiona l part***	$f(x)=x-[x]$		$\mathbb{R}$	$[0,1)$
l)	***Square root***	$f(x)=\sqrt{x}$		$[0,\infty)$	$[0,\infty)$
m)	***Cube root***	$f(x)=\sqrt[3]{x}$		$\mathbb{R}$	$\mathbb{R}$

n)	***Signum function***	$y=\operatorname{sgn}(x)=\begin{cases}\frac{\lvert x\rvert}{x}, x\neq 0,\\ 0, x=0\end{cases}$		$\mathbb{R}$	{- 1, 0,1}

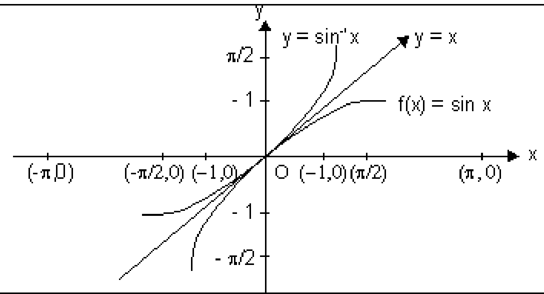
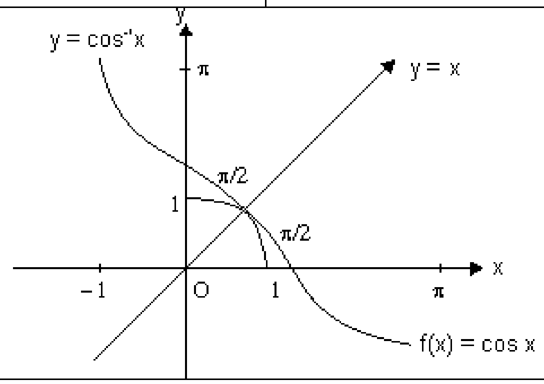
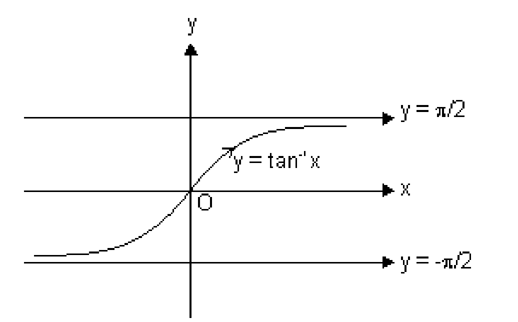
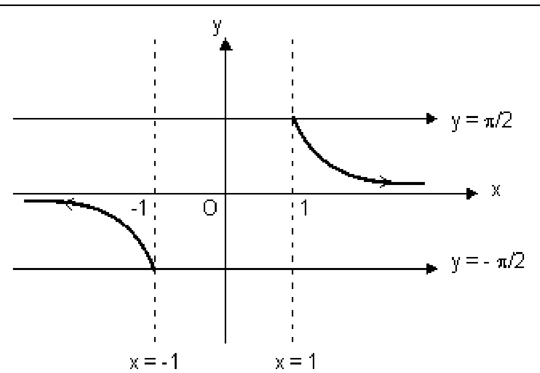
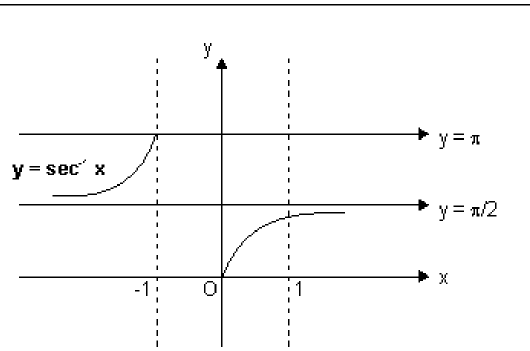
TRIGNOMETRIC FUNCTIONS

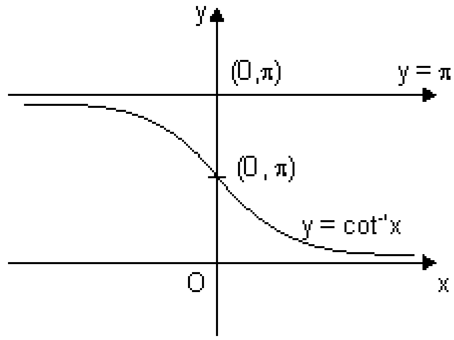
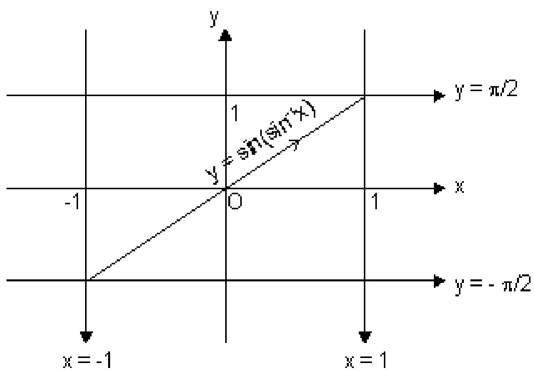
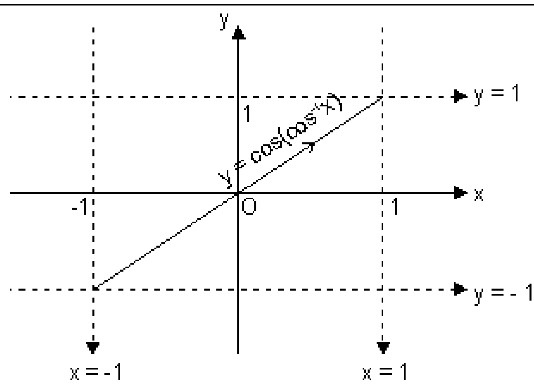
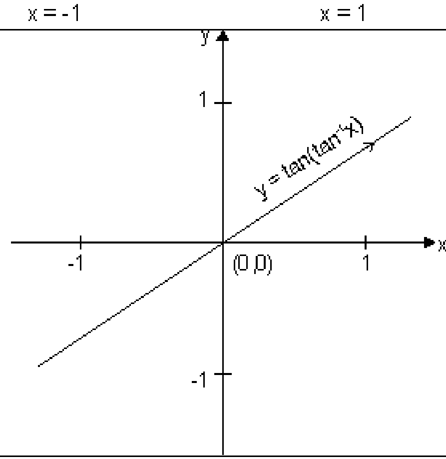
	FUNCTION	PERIOD	GRAPH	DOMAIN	RANGE
a)	***sinx***	2π		$\mathbb{R}$	[-1,1]
b)	***cosx***	2π		$\mathbb{R}$	[-1,1]
c)	***tanx***	π		$\mathbb{R}-\left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R}$

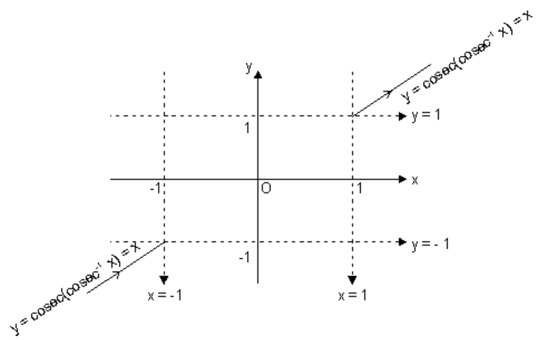
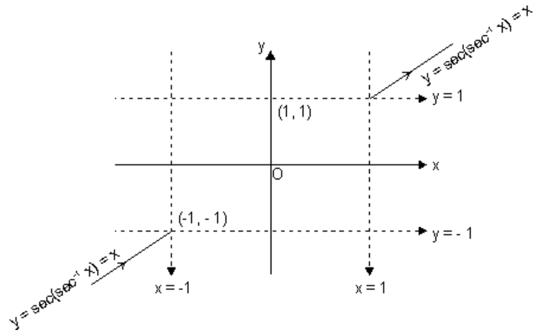
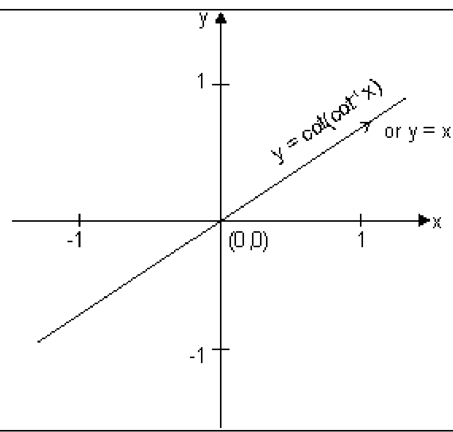
d)	***cosecx***	2π		$\mathbb{R}-\{n\pi : n\in\mathbb{Z}\}$	$\mathbb{R}-(-1,1)$
e)	***secx***	2π		$\mathbb{R}-\left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R}-(-1,1)$
f)	***cotx***	π		$\mathbb{R}-\{n\pi : n\in\mathbb{Z}\}$	$\mathbb{R}$

INVERSE TRIGNOMETRIC FUNCTIONS

	FUNCTION	GRAPH	DOMAIN	RANGE

a)	$\sin^{-1} x$		[-1, 1]	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$
b)	$\cos^{-1} x$		[-1, 1]	$\left[0,\pi\right]$
c)	$\tan^{-1} x$		$\mathbb{R}$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$
d)	$\cos ec^{-1}x$		$\mathbb{R}-(-1,1)$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$
e)	$\sec^{-1} x$		$\mathbb{R}-(-1,1)$	$\left[0,\pi\right]-\left\{\frac{\pi}{2}\right\}$

f)	$\cot^{-1} x$		$\mathbb{R}$	$(0,\pi)$
g)	**sin(sin^{-1}x)**		[-1, 1]	[-1, 1]
h)	**cos(cos^{-1}x)**		[-1, 1]	[-1, 1]
i)	**tan(tan^{-1}x)**		$\mathbb{R}$	$\mathbb{R}$

j)	**cosec($cosec^{-1}$x)**		$\mathbb{R}-(-1,1)$	$\mathbb{R}-(-1,1)$
k)	**sec(sec^{-1}x)**		$\mathbb{R}-(-1,1)$	$\mathbb{R}-(-1,1)$
l)	**cot(cot^{-1}x)**		$\mathbb{R}$	$\mathbb{R}$

1.3 PROPERTIES,DOMAIN AND RANGE

Properties of some standard functions:

1. **<u>Properties of modulus function</u>**:

(i) $|x| \leq a \Rightarrow -a \leq x \leq a \quad (a \geq 0)$

(ii) $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a \quad (a \geq\ )$

(iii) $|x+y| \leq |x| + |y|$

(iv) $|x+y| \geq \left||x|-|y|\right|$

2. **Properties of greatest integer function**:

(i) $[x] = x$ holds if $x \in I$ (I is Integer).

(ii) $[x + I] = [x] + I$ if I is integer.

(iii) $[x + y] \geq [x] + [y]$

(iv) If $[\phi(x)] \geq I$, then $\phi(x) \geq I$.

(v) If $[\phi(x)] \leq I$, then $\phi(x) < I + 1$

(vi) If $[x] > n \Rightarrow x \geq n + 1, n \in I$.

(vii) $[-x] = -[x]$ if $x \in I$
& $[-x] = -[x] - 1$ if $x \notin I$.

(viii) $[x + y] = [x] + [x + y - [x]]$ for all $x, y \in R$.

(ix) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \ldots\ldots\ldots\ldots + \left[x + \frac{n-1}{n}\right]$
$= [n x], n \in N$ $N \rightarrow$ natural no.

3. **Properties of fraction part of x**

(i) $\{x\} = x$ if $0 \leq x < 1$

(ii) $\{x\} = 0$ if $x \in I$

(iii) $\{-x\} = 1 - \{x\}$ if $x \notin I$

4. **Properties of logarithmic function:**

(i) $\log_a a = 1$ $(a>0, a\neq1)$

(ii) $\log_e(ab) = \log_e a + \log_e b$ $(a,b>0)$

(iii) $\log_e(a/b) = \log_e a - \log_e b$ $(a,b>0)$

(iv) $\log_e a^m = m \log_e a$ $(a>0, m\in R)$

(v) $\log_b a = 1/(\log_a b)$ $(a,b>0$ and $a,b\neq1)$

(vi) $\log_b a = \log_m a / \log_m a$ $(a,b,m>0 \neq1)$

(vii) $a^{\log_a m} = m$ $(a>0, a\neq1; m>0)$

(viii) $\log_m x > \log_m y \Rightarrow \begin{Bmatrix} x > y; m > 1 \\ x < y; 0 < m < 1 \end{Bmatrix}$

(ix) $\log_m a > b \Rightarrow \begin{Bmatrix} a > m^b; m > 1 \\ a < m^b; 0 < m < 1 \end{Bmatrix}$

(x) $\log_m a < b \Rightarrow \begin{Bmatrix} a < m^b; m > 1 \\ a > m^b; 0 < m < 1 \end{Bmatrix}$

(xi) $\log_m a = b \Rightarrow a = m^b$ (m,a>0;m≠1b∈R)

Exercise1.2:

Q1: Solve the following equations:

I. $x^2 + 3x + 2 > 0$

II. $x^2 + 3x + 5 < 0$

III. $x^2 + 2x + 1 > 0$

IV. $\dfrac{(x-1)(x-2)}{(x-3)(x+1)} > 0$

V. $1 \le |x-1| \le 3$

VI. $\dfrac{2x}{2x^2 + 5x + 2} > \dfrac{1}{x+1}$

VII. $\left| \dfrac{x^2 - 5x + 4}{x^2 - 4} \right| \le 1$

VIII. $\log_{10}(x^2 - 3x + 3) \ge 0$

IX. $\dfrac{(2x-1)(x-1)^2(x-2)^2}{(x-4)^4} > 0$

X. $\dfrac{|x|-1}{|x|-2} \ge 0, x \in \mathbb{R}, x \ne \pm 2$

XI. $\dfrac{|x+3| + x}{x+2} > 1$

XII. $|x+1| + \sqrt{x-1} = 0$

XIII. $\dfrac{x^2 - 2x + 2^{|a|}}{x^2 - a^2} > 0$

XIV. $\log_3(5 + 4\log_3(x-1)) = 2$

XV. $\log_2(3-x) + \log_2(1-x) = 3$

XVI. $\log_{\frac{1}{5}} \dfrac{4x+6}{x} \ge 0$

XVII. $\log_{\frac{x+4}{2}} \log_2 \dfrac{2x-1}{3+x} < 0$

XVIII. $4\{x\} = x + [x]$

Exercise1.3:

Q1: Find the domains of the following functions, assuming the functions to be real for given $f(x)=$

I. $\sqrt{1-x^2}$

II. $\frac{1}{\sqrt{1-x^2}}$

III. $\frac{1}{1+\sin x}$

IV. $\sqrt{2-x}+\sqrt{3+x}$

V. $\log_e\left|x^2-3x+2\right|$

VI. $\sqrt{\sin x-1}$

VII. $\frac{1}{[x]}$

VIII. $\log[x]$

IX. $\sin\sqrt{x}$

X. $\frac{1}{\sqrt{[x]^2-3[x]+2}}$

XI. log$_2$log$_3$log$_4$log$_5$x

XII. $\sqrt{x^2-1}+\frac{1}{\sqrt{x^2-5x+4}}$

XIII. $\sin\left(\ln\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right)$

XIV. $\frac{1}{\sqrt{[x]-x}}$

XV. $\frac{\sqrt{1}}{\sqrt{|x|-x}}$

XVI. $\log\left(\log_{|\sin x|}(x^2-8x+23)-\frac{3}{\log_2|\sin x|}\right)$

XVII. $\sqrt{\log_{.4}\left(\frac{x-1}{x+5}\right)}$

XVIII. $\cos^{-1}\left(\frac{1-2|x|}{3}\right)+\log_{|x-1|}x$

XIX. $[\sin x]\cos\left(\frac{\pi}{[x-1]}\right)$

XX. $\frac{1}{\sqrt{\log_{1/2}(x^2-7x+13)}}$

Q2: Find the domain of single valued function y = f(x) given by the equation $10^x+10^y=10$

Exercise1.4:

Q1: Find the range of the following functions, assuming real functions given by $f(x)=$

I. x

II. x^2+1

III. $\sqrt{\sin x}$

IV. $[\sin x]$

V. $\dfrac{1}{[x]}$

VI. $\dfrac{1}{x+1}$

VII. $\left[\dfrac{1}{x}\right], x>0$

VIII. $[|\cos x|]$

IX. $|[\cos x]|$

X. x^2+3x+2

XI. $\dfrac{x^2-3x+2}{x^2+x-6}$

XII. $\dfrac{x^2+x-1}{x^2-x+2}$

XIII. $\dfrac{1}{2+\sin 3x+\cos 3x}$

XIV. $[x^2]-[x]^2$

XV. x^3+3x^2+4x+5

XVI. $\dfrac{x-[x]}{1-[x]+x}$

XVII. $\log_e(3x^2-4x+5)$

XVIII. $\log_3(\log_{1/2}(x^2+4x+4))$

1.4 ANALYSIS OF A FUNCTION

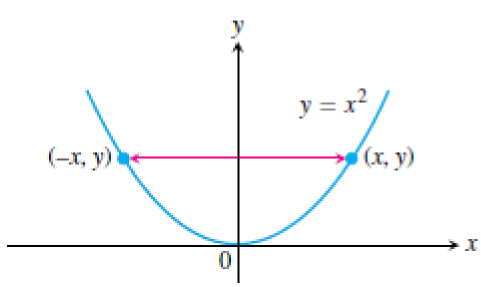
Even and odd functions

Definition: A function y=f(x) is an

Even function of x if $f(-x)=f(x)$

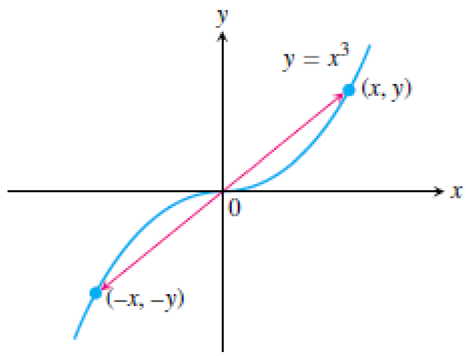
Odd function of x if $f(-x)=-f(x)$

Graphs of even function are symmetric about y axis ,example $y=x^2$



(a)

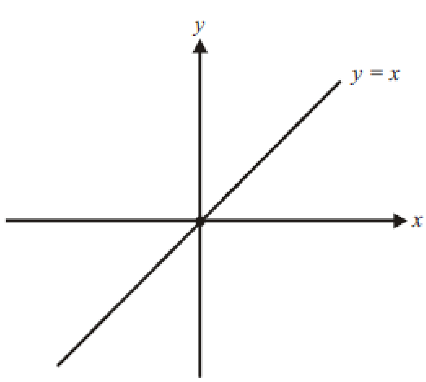
Graphs of odd function are symmetric about origin ,example $y = x^3$



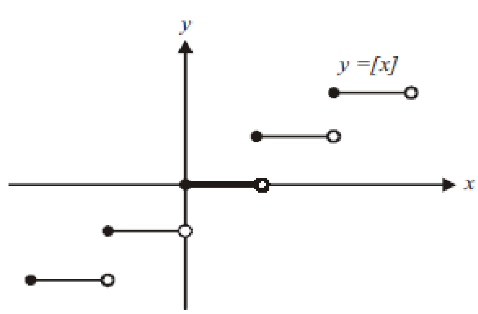
Increasing and decreasing function:

If the graph of a function *climbs* or *rises* as you move from left to right, we say that the function is *increasing*. If the graph *descends* or *falls* as you move from left to right, the function is *decreasing*

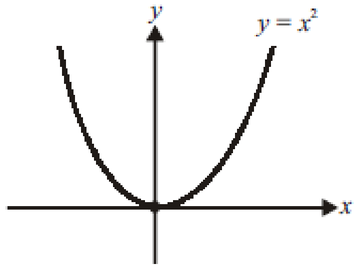
If	$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$	function is strictly increasing
If	$x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$	function is increasing
If	$x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$	function is strictly deccreasing
If	$x_1 > x_2 \Rightarrow f(x_1) \ge f(x_2)$	function is decreasing



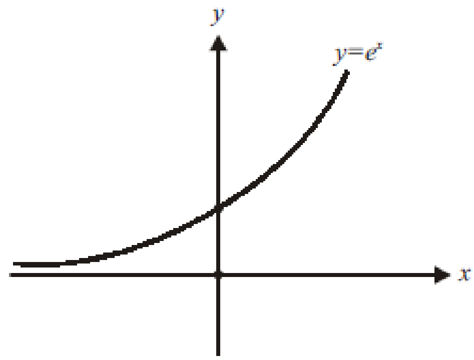
strictly increasing



increasing



strictly decreasing on $(-\infty,0]$
strictly increasing on $[0, \infty)$



strictly increasing on $[-\infty, 0)$

Periodic function:

A function $f(x)$ is **periodic** if there is a positive number T such that $f(x+T) = f(x)$ for every value of x. The smallest such value of T is the **period** of f.
For example, sin x has periods 2π , 4π ,6π , -4π , $2n\pi$ etc. but the fundamental period is 2π.

Properties of periodic function:

(i) If f(x) is periodic with period T, then, c.f(x), f(x + c) & f(x) ± c is periodic with period T.
eg. if sin x has period 2π . Then f(x) = 5 sinx + 4 is also periodic with period 2π .

(ii) If f(x) is periodic with period T, then, kf(cx + d) has period $\frac{T}{|C|}$

f(x) = 7 sin2x - 12 has period $\frac{2\pi}{|2|} = \pi$ as sinx is periodic with period .

(iii) If $f_1(x)$, $f_2(x)$ are periodic functions with periods T_1, T_2 respectively then; $h(x) = af_1(x) \pm bf_2(x)$ has period as

$$= \begin{cases} \text{LCm of } \{T_1, T_2\}; \text{ If } h(x) \text{ is not an even function} \\ \qquad\qquad\qquad \text{or} \\ \frac{1}{2} \text{ LCM of } \{T_1, T_2\}, f_1(x) \; \& f_2(x) \text{ are complementar pair wise comparable functions} \end{cases}$$

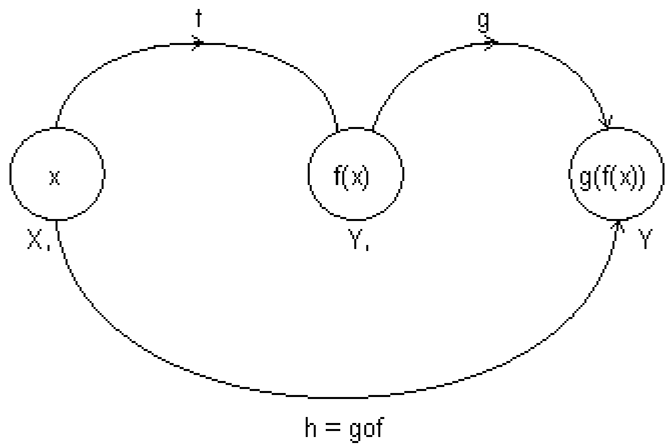
LCM of $\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM of } (a, b, c)}{\text{HCF of } (b, d, f)}$

Note: LCM of rational and irrational is not possible. eg. 2π . is irrational & 1 is rational.

Composite Function:

Let two function,

f : x $\rightarrow$ Y_1 and   g : Y_1 $\rightarrow$ Y



Let another function h : X $\rightarrow$ Y

Such that

h(x) = g(f(x)) = (gof)(x)

domain (gof) = {x : x $\in$ domain (f), f(x) $\in$ domain (a)}

domain fog(x) = {x : x $\in$ domain (g), g(x) $\in$ domain f}

gof exist iff range of f $\subseteq$domain of g

fog exist iff range of g $\subseteq$domain of f.

<u>Properties of composite function</u>:

(i) f is even, g is even $\Rightarrow$ fog is even function.

(ii) f is odd, g is odd $\Rightarrow$ fog is odd function.

(iii) f is even, g is odd $\Rightarrow$ fog is even function.

(iv) f is odd, g is even $\Rightarrow$ fog is even function.

<u>Exercise1.5:</u>

Q1: Check weather the following functions $f(x)$ are even or odd

I. $\sin x - \cos x$

II. $\frac{x}{e^x - 1} + \frac{x}{2} + 1$

III. $\sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2}$

IV. $\dfrac{x(\sin x+\tan x)}{\left[\dfrac{x+\pi}{\pi}\right]-\dfrac{1}{2}}$

Q2: If the function f satisfies $f(x+y)+f(x-y)=2f(x).f(y)\ \forall\ x, y\in R$, and $f(0)\neq 0$, prove that $f(x)$ is even.

Q3: If f is non-periodic and g is periodic, what can you say about $f($

$g(x))$ and $g(f(x))$?

Q4: What are the periods of the following functions $f(x)=$

I. $|\sin 2x|+|\sin 3x|$

II. $\sin^4 x+\cos^4 x$

III. $2\cos\left(\dfrac{x-\pi}{3}\right)$

IV. $\sin^2 \pi x$

Q5: Prove that if f is increasing, f^{-1} is increasing and if f is decreasing, f^{-1} is decreasing

Q6: If $f(x)=\sin x+\cos x$ and $g(x)=x^2-1$ then find the interval in which g(f(x) is invertible?

Q7: If $f(x+y)=f(x).f(y)\ \forall$ real *x,y* and $f(0)\neq 0$ then prove that the function $f(x)=\dfrac{f(x)}{1+(f(x))^2}$ is an even function.

Q8: If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\dfrac{1}{x}$,then find $f^{-1}(x)$ (assuming bijective)

Q9: If $f:[1/2,\infty)\to[3/4,\infty)$ is given by $f(x)=x^2-x+1$,then find $f^{-1}(x)$ (assuming bijective)

Q10: Find the inverse of the following functions (assuming bijective)

I. $f(x)=\sin^{-1}(x/3), x\in[-3,3]$

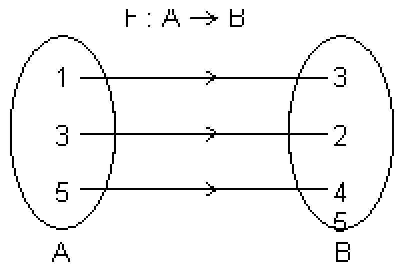
II. $f(x)=\ln(x^2+3x+1), x\in[1,3]$

Q11: If $f(x)=\begin{cases} x-1; -1\le x\le 0 \\ x^2; 0\le x\le 1\end{cases}$ and $g(x)=\sin x$. Then find $h(x)=f\,|g(x)|+|f(g(x))|$

Mapping of a Function:

Mapping of function: A function exists only if," to every element in domain there exists unique image in co domain.
f : A $\rightarrow$ B

One - One Mapping (Injective): A function f: A $\rightarrow$ B is said to bwe one-one mapping if different elements of A have different images in B. $\Rightarrow$ no two elements of set A can have same 'f' image
eg.



f(x) is one-one

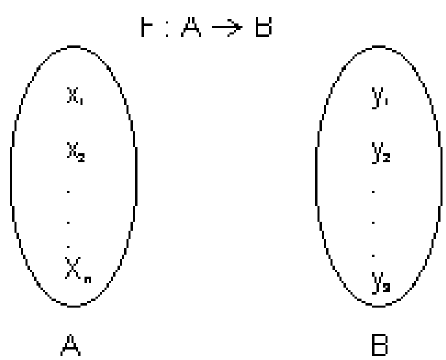
Methods to find one-one mapping:

(1) **Graphically**: A fun. is one-one if no line 11 to x-axis meets graph of function at more than one point.

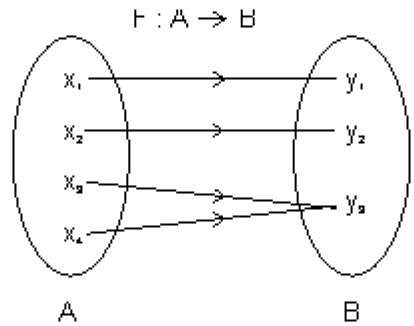
By Calculus: If f(x) is function and if f'(x) $\geq$ 0 $\forall$ x $\in$ domain i.e. increasing or f'(x) $\leq$ 0 $\forall$ x $\in$ domain i.e. decreasing, then one-one.

No. of one-one Mapping: If A & B are finite sets having m & h elements, then, no. of one-one function from A to B

$$= \begin{cases} {}^{n}P_m & \text{if } n \geq m \\ 0 & \text{if } n < m \end{cases}$$



Many one Mapping: A mapping f : A $\rightarrow$ B is many one if two or more elements of set A have the same image in B.

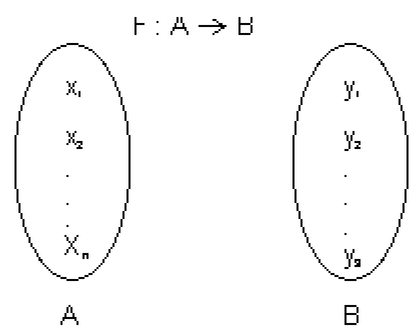


i.e. $f(x^3) = f(x_4) = y_3$

<u>Onto Function (Surjective)</u>: If function $f : A \rightarrow B$ is such that each element of B is the 'f' image of at least one element in A.
i.e. range of f = co domain $\Rightarrow f(A) = B$

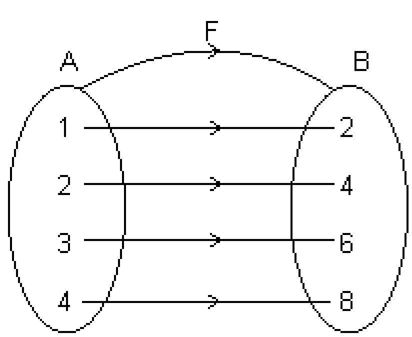
<u>Into Function</u>: A function of $f : A \rightarrow B$ is into if there exists an element in B having no pre image in A.

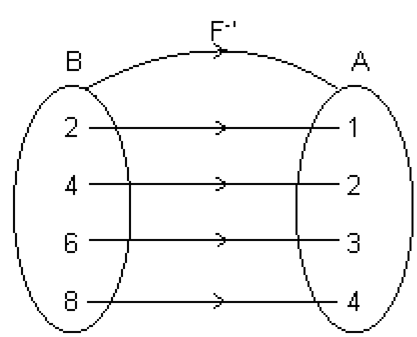
<u>No. of one-one onto mapping (bijection)</u>: If A & B are finite sets & $f : A \rightarrow B$ is a bijection.
Then A & B have same no. of elements.



Total no. of bijection $A \rightarrow B = n\,!$

<u>Inverse function</u>: Let $f : A \rightarrow B$ be one-one & onto function then there exists a unique function.





$g : B \rightarrow A$ such that $f(x) = y \Leftrightarrow g(y) = x, \forall x \in A \;\&\; y \in B$
Then g is c/d inverse of F.
So, $g = f^{-1} : B \rightarrow A = \{(f(x), x) \mid (x, f(x) \in f\}$

Domain of f = {1, 2, 3, 4} = range of f^{-1}
Range of f = {2, 4, 6, 8} = domain of f^{-1}.

Note: (1) Graph of y = f(x) and y = f^{-1}(x) are mirror images to each other.

(2) If f : A $\rightarrow$ B & g : B $\rightarrow$ C are two bijections, then gof : A $\rightarrow$ C is bijection & $(gof)^{-1} = (f^{-1}og^{-1})$.

(3) If fog = gof then either f^{-1} = g or g^{-1} = f and (fog)(x) = (gof)(x) = (x)

Exercise1.6:

Q1: Is $f(x)=\dfrac{x+1}{x+2}$ one one or many one?

Q2: Is $f(x)=\dfrac{x^2-8x+18}{x^2+4x+30}$ one one or many one?

Q3: Are the following functions into or onto?

I. $f:\mathbb{R}\rightarrow\mathbb{R}$ $f(x)=\sin x$

II. $f:[-1,1]\rightarrow[0,1]$ $f(x)=x^2$

III. $f:(0,\infty)\rightarrow\mathbb{R}$ $f(x)=\ln x$

IV. $f:(1,\infty)\rightarrow\mathbb{R}$ $f(x)=\ln x$

Exercise level 1:

1. The period of the function, $f(x) = [\sin 3x] + |\cos 6x|$ is : ([.] denotes the greatest integer less than or equal to x)

(a) π (b) $\frac{2\pi}{3}$ (c) 2π (d) $\frac{\pi}{2}$

---

2. The function $f(x) = \frac{\sec^{-1} x}{\sqrt{x - [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x is defined for all x belonging to :

(a) R (b) R - {(-1, 1) ∪ {n : n ∈ I)} (c) R′ - (0, 1) (d) R′ - {n : n ∈ N}

---

3. The function $f(x) = \sqrt{\log_{x^2}(x)}$ is defined for x belonging to :

(a) $(-\infty, 0)$ (b) $(1, \infty)$ (c) $(0, \infty)$ (d) none of these

---

4. If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x . \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then $(gof)(x) =$

(a) 1 (b) -1 (c) x (d) none of these

---

5. Let [x] denote the greatest integer $\leq x$. The domain of definition of function $f(x) = \sqrt{\frac{4 - x^2}{[x] + 2}}$ is

(a) $(-\infty, -2) \cup [-1, 2]$ (b) [0, 2] (c) [-1, 2] (d) (0, 2)

---

6. The domain of definition of the function $f(x) = \sqrt{\log_{10}\left(\frac{5x - x^2}{4}\right)}$ is

(a) [1, 4] (b) (1, 4) (c) (0, 5) (d) [0, 5]

---

7. The range of the function $f(x) = \frac{1}{2 - \cos 3x}$ is

(a) $\left[-\frac{1}{3}, 0\right]$ (b) R (c) $\left[\frac{1}{3}, 1\right]$ (d) none of these

---

8. The domain of definition of the function $f(x) = \frac{1}{\sqrt{|x| - x}}$ is

(a) R (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) none of these

---

9. The function $f(x) = \log(x + \sqrt{x^2+1})$ is

(a) an even function (b) an odd function (c) periodic function (d) none of these

10. The function $f(x) = \cos\left(\log(x + \sqrt{x^2+1})\right)$ is

(a) even (b) odd (c) constant (d) none of these

11. Thr period of the function $f(x) = \sin^4 x + \cos^4 x$ is

(a) π (b) $\pi/2$ (c) 2π (d) none of these

12. Which of the following functions is an odd function

(a) f(x) = constant (b) $f(x) = \sin x + \cos x$ (c) $f(x) = \sin\left(\log(x + \sqrt{x^2+1})\right)$ (d) $f(x) = 1 + x + 2x^3$

13. The domain of definition of the function $f(x) = {}^{7-x}P_{x-3}$ is

(a) [3, 7] (b) {3,4,5,6,7} (c) {3,4,5} (d) none of these

14. The function $f(x) = \dfrac{\sin^4 x + \cos^4 x}{x + x^2 \tan x}$ is

(a) even (b) odd (c) periodic with period π (d) periodic with period 2π

15. Range of $f(x) = |x| + |x+1|$ is

(a) $[0, \infty)$ (b) $(0, \infty)$ (c) $(1, \infty)$ (d) $[1, \infty)$

16. Range of $\tan^{-1}x - \cot^{-1}x$ is

(a) $(0, \pi)$ (b) $\left[\dfrac{-3\pi}{2}, \dfrac{\pi}{2}\right]$ (c) $\left(\dfrac{2}{3}, 1\right)$ (d) $\left[\dfrac{2}{3}, 1\right]$

17. The range of $\sqrt{|x| - x}$ is

(a) $(0, \infty)$ (b) $[0, \infty)$ (c) $(-\infty, 0)$ (d) $(-\infty, 0]$

18. If $f(x) = x^2\left(\dfrac{10^{2x}-1}{10^{2x}+1}\right)$ then 'f' is

(a) an even function (b) an odd function (c) neither even nor odd (d) cannot be determined

19. The domain of the function $f(x) = \dfrac{1}{\log_{10}(1-x)} + \sqrt{(x+2)}$ is

(a) [-3, -2] excluding (-2.5) (b) [0, 1] excluding 0.5

(c) [-2, 1], excluding 0 (d) None of these

20. The period of $e^{\cos^4 \pi x + x - [x] + \cos^2 \pi x}$ is ([.] denotes the greatest integer function)

(a) 2 (b) 1 (c) 0 (d) -1

21. The domain of the function $f(x) = \sqrt{\sin^{-1}(\log_2 x)} + \sqrt{\cos(\sin x)} + \sin^{-1}\left(\dfrac{1+x^2}{2x}\right)$

(a) $\{x : 1 \le x \le 2\}$ (b) {1} (c) Not defined for any value x (d) {-1, 1}

22. If [.] denotes the greatest integer function then the domain of the real valued function $\log_{[x+1/2]}|x^2-x-2|$ is

(a) $\left[\frac{3}{2},\infty\right)$ (b) $\left[\frac{3}{2},2\right)\cup(2,\infty)$ (c) $\left(\frac{1}{2},2\right)\cup(2,\infty)$ (d) None of these

23. The domain of the function $f(x)=\sqrt[6]{4^x+8^{\frac{2}{3}(x-2)}-52-2^{2(x-1)}}$ is

(a) (0, 1) (b) $[3, \infty)$ (c) (1, 0) (d) none

24. A function whose graph is symmetrical about the y-axis is given by

(a) $f(x)=\log_e(x+\sqrt{x^2+1})$ (b) $f(x+y)=f(x)+f(y)$ for all $x, y \in R$

(c) $f(x)=\cos x+\sin x$ (d) none of these

25. If $f(x)=x^2-x+1,\ x\in\left[\frac{1}{2},\infty\right)$ then value of 'x' satisfying $f(x)=f^{-1}(x)$ is

(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) none of these

26. If $f(x)=\sqrt{\log_{0.4}\left(\frac{x-1}{x-5}\right)}$ and $g(x)=x^2-36$, then $D_{f/g}$ is

(a) $(-\infty, 0)\sim\{-6\}$ (b) $(0, \infty)\sim\{1, 6\}$

(c) $(1, \infty)\sim\{6\}$ (d) $[1, \infty)\sim\{6\}$

27. The domain of the real-valued function $f(x)=\log_e|\log_e x|$ is

(a) $(0,1)\cup(1,\infty)$ (b) $(0,\infty)$ (c) (e,∞) (d) $(1,\infty)$

28. If f(x) and g(x) are two functions of 'x' such that $f(x)+g(x)=e^x$ and $f(x)-g(x)=e^{-x}$, then

(a) f(x) is odd, g(x) is odd (b) f(x) is even, g(x) is even

(c) f(x) is even, g(x) is odd (d) f(x) is odd, g(x) is even

29. The inverse of the function $y=\log_a(x+\sqrt{x^2+1})$ $(a>0,\ a\neq 1)$ is

(a) $\frac{1}{2}(a^x-a^{-x})$ (b) not defined for all x

(c) defined for only positive x (d) none of these

30. Let $f(x)=\cos\sqrt{p}\,x$, where $p=[a]$ = the greatest integer less than or equal to 'a'. If the period of f(x) is π, then

(a) $a\in[4,5]$ (b) $a=[4,5]$ (c) $a\in[4,5)$ (d) none of these

31. If $f(x)=e^{x-[x]+|\cos\pi x|+|\cos 2\pi x|+........+|\cos n\pi x|}$, then period of f(x) is

(a) 1 (b) $\frac{1}{n}$ (c) $\frac{n^2-1}{n^2+1}$ (d) $\frac{1}{1.2.3......n}$

32. Let $f(x)=\sin^{-1}\left(\frac{2-|x|}{4}\right)+\cos^{-1}\left(\frac{2-|x|}{4}\right)+\tan^{-1}\left(\frac{2-|x|}{4}\right)$, then D_f is

(a) [-6, 6] (b) $[6, \infty)$ (c) [-6, 3] (d) [-3, 6]

---

33. Identify the statement(s) which is/are incorrect ?

(a) The function $f(x) = \cos(\cos^{-1} x)$ is neither odd nor even

(b) The fundamental period of $f(x)=\cos(\sin x)+\cos(\cos x)$ is π

(c) The range of the function $f(x) = \cos(3 \sin x)$ is $[-1, 1]$

(d) None of these

---

34. Let $f(x)=4\cos\sqrt{x^2-\frac{\pi^2}{9}}$. Then

(a) $D_f=\left[\frac{\pi}{3}, \infty,\right), R_f=[-1, 1]$ (b) $D_f=\left[\frac{\pi}{3}, \infty,\right), R_f=[-2, 2]$

(c) $D_f=\left(-\infty, -\frac{\pi}{3}\right]\cup\left[\frac{\pi}{3}, \infty\right)$ and $R_f=[-4, 4]$ (d) $D_f=\left(-\infty, -\frac{\pi}{3}\right), R_f=(0, 4]$

---

35. The range of the function $f(x)=\sin(xe^{[x]}+x^2-x),\ x\in(-1, \infty)$ where [x] denotes the greatest integer function is:

(a) ϕ (b) [0, 1] (c) [-1, 1] (d) R

---

36. The domain of $f(x) = \log_2 \log_3 \log_{4/\pi} (\tan^{-1} x)^{-1} =$

(a) R (b) $(4/\pi, \infty)$ (c) (0, 1) (d) None of these

---

37. If f(x) is a periodic function of the period 'λ' then $f(\lambda x + a)$, where a is a constant, is a periodic function of the period

(a) 1 (b) λ (c) $\frac{\lambda}{a}$ (d) none of these

---

38. If $f(x)=\cos^{-1}(x-x^2)+\sqrt{\left(1-\frac{1}{|x|}\right)}+\frac{1}{[x^2-1]}$ then domain of f(x) is (where [.] is the greatest integer)

(a) $\left[\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ (b) $\left(\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ (c) $\left(-\sqrt{2}, \frac{1-\sqrt{2}}{2}\right)$ (d) none of these

---

39. Let $f(x) = \sin x$, $g(x) = \ln|x|$. If the ranges of the composite functions fog and gof are R_1 and R_2 respectively, then

(a) $R_1=(-1,1), R_2=(-\infty,0)$ (b) $R_1=(-\infty,0], R_2=[-1,1]$

(c) $R_1=(-1,1), R_2=(-\infty,0]$ (d) $R_1=[-1,1], R_2=(-\infty,0]$

---

40. If $f(x)=\log\frac{1+x}{1-x}$, then

(a) f(x) is even (b) $f(x_1).f(x_2)=f(x_1+x_2)$ (c) $\frac{f(x_1)}{f(x_2)}=f(x_1-x_2)$ (d) f(x) is odd

Exercise level 2:

1. The domain of the function $f(x)=\log(1-x)+\sqrt{x^2-1}$ is

(a) [-1, 1] (b) $(1, \infty)$ (c) (0, 1) (d) $(-\infty, -1]$

2. The range of the function $f(x)=\dfrac{1+x^2}{x^2}$ is equal to

(a) [0, 1] (b) (0, 1) (c) $(1, \infty)$ (d) $[1, \infty)$

3. The curves $y=|x|^3+3|x|^2+2$ and $y=x^3+3x^2+2$ have the same graph for

(a) $x>0$ (b) $x\geq 0$ (c) all x except 0 (d) all x

4. Domain of the function $f(x)=\dfrac{1}{x^2}+2^{\sin^{-1}x}+\dfrac{1}{\sqrt{x-3}}+7$ is

(a) ϕ (b) R - {0} (c) R (d) None of these

5. The domain of definition of the function $y=3e^{\sqrt{x^2-1}}\log(x-1)$ is

(a) $(1, \infty)$ (b) $[1, \infty)$ (c) $R \sim \{1\}$ (d) $(-\infty, -1)\cup(1, \infty)$

6. The range of the function f(x) = cos [x], where $-\dfrac{\pi}{2}<x<\dfrac{\pi}{2}$, is

(a) {-1, 1, 0} (b) {cos 1, 1, cos 2} (c) $\{\cos 1, -\cos 1, 1\}$ (d) none of these

7. If $b^2 - 4ac = 0$ and $a>0$, then domain of the function $y=\log(ax^3+(a+b)x^2+(b+c)x+c)$ is

(a) $R\sim\left\{-\dfrac{b^2}{2a}\right\}$ (b) $R\sim\left(\left\{-\dfrac{b}{2a}\right\}\cup\{x\,|\,x\geq -1\}\right)$

(c) $R\sim\left(\left\{-\dfrac{b}{2a}\right\}\cup(-\infty, -1]\right)$ (d) none of these

8. Which of the following functions is an even function?

(a) $f(x)=\dfrac{a^x+a^{-x}}{a^x-a^{-x}}$ (b) $f(x)=\dfrac{a^x+1}{a^x-1}$ (c) $f(x)=x\dfrac{a^x-1}{a^x+1}$ (d) $f(x)=\log_2\left(x+\sqrt{x^2+1}\right)$

9. If $(\log_3 x)(\log_x 2x)(\log_{2x} y)=\log_x x^2$, then y is equal to

(a) 9 (b) 18 (c) 27 (d) 81

10. The range of the function $f(x)=\dfrac{x^2-x+1}{x^2+x+1}$ is

(a) R (b) $[3, \infty)$ (c) $\left[\dfrac{1}{3}, 3\right]$ (d) none of these

11. The domain of the function $f(x)=\sqrt{2-2x-x^2}$ is

(a) $-3 \le x \le \sqrt{3}$ (b) $-1-\sqrt{3} \le x \le -1+\sqrt{3}$ (c) $-2 \le x \le 2$ (d) none of these

12. If the domain of the function $f(x)=x^2-6x+7$ is $(-\infty, \infty)$, then range of the function is

(a) $(-\infty, \infty)$ (b) $[-2, \infty)$ (c) (-2, 3) (d) $(-\infty, -2)$

13. The domain of the function $f(x)=\sin^{-1}\left(\log_2 \dfrac{x^2}{2}\right)$ is

(a) $-1 \le x \le 1$ (b) $0 \le x \le 1$ (c) $1 \le x \le 2$ (d) none of these

14. If f(x) = cos (log x) then $f(x)f(y)-\dfrac{1}{2}\left(f\left(\dfrac{x}{y}\right)+f(xy)\right)$ has the value

(a) 1 (b) $\dfrac{1}{2}$ (c) -2 (d) 0

15. The inverse of the function $f(x)=\dfrac{e^x-e^{-x}}{e^x+e^{-x}}+2$ is given by

(a) $\log_e\left(\dfrac{x-2}{x-1}\right)^{1/2}$ (b) $\log_e\left(\dfrac{x-1}{3-x}\right)^{1/3}$ (c) $\log_e\left(\dfrac{x}{2-x}\right)^{1/2}$ (d) $\log_e\left(\dfrac{x-1}{x+1}\right)^{-2}$

16. The range of the function for real x of $y=\dfrac{1}{2-\sin 3x}$ is

(a) $\dfrac{1}{3} \le y \le 1$ (b) $-\dfrac{1}{3} \le y < 1$ (c) $-\dfrac{1}{3} > y > 1$ (d) $\dfrac{1}{3} > y > 1$

17. The domain of the function $\dfrac{1}{[x]}+\sqrt{2x-x^2}$, where [.] denotes greatest integer function.

(a) [1, 2] (b) [0, 2] (c) [0, 1) (d) [1, 2)

18. Domain of $\sin^{-1}(\log_3(x/3))$ is

(a) [1, 9] (b) [-1, 9] (c) [-9, 1] (d) [-9, -1]

19. The range of $f(x) = {}^{7-x}P_{x-3}$ is

(a) {1, 2, 3} (b) {1, 2, 3, 4, 5, 6} (c) {1, 2, 3, 4} (d) {1, 2, 3, 4, 5}

20. The value of $n \in Z$ for which the function $f(x) = \dfrac{\sin nx}{\sin\left(\dfrac{x}{n}\right)}$ has 4π as its period is

(a) 2 (b) 3 (c) 5 (d) 4

21. The domain of the function $f(x) = \log_2(\log_3(\log_4 x))$ is

(a) $x < 4$ (b) $x > 4$ (c) $0 < x < 2$ (d) $2 < x < 4$

22. If f(x) is an odd periodic with period 2, then f(4) is

(a) 0 (b) 2 (c) 4 (d) -4

23. If $f(x) = 1 + \alpha x$, ($\alpha \neq 0$) is the inverse of itself then the value of α is

(a) -2 (b) -1 (c) 0 (d) 2

24. If g(x) be a function defined on [-1, 1] and if the area of the equilateral triangle with two of its vertices at (0, 0) and (x, g(x))) is $\dfrac{\sqrt{3}}{4}$, then the function is

(a) $g(x) = \pm\sqrt{1-x^2}$ (b) $g(x) = -\sqrt{1-x^2}$ (c) $g(x) = \sqrt{1-x^2}$ (d) $g(x) = \sqrt{1+x^2}$

25. Which of the following functions is periodic

(a) $f(x) = x - [x]$ (b) $f(x) = \begin{cases} x\sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (c) $f(x) = x\cos x$ (d) none of these

26. $f(x) = \sqrt{\dfrac{(x+1)(x-3)}{x-2}}$ is real valued in the domain

(a) $(-\infty, -1] \cup [3, \infty)$ (b) $(-\infty, -1] \cup (2, 3]$ (c) $[-1, 2) \cup [3, \infty)$ (d) none of these

27. The domain of definition of the function y(x) given by the equation $2^x + 2^y = 2$ is

(a) $0 < x \le 1$ (b) $0 \le x \le 1$ (c) $-\infty < x \le 0$ (d) $-\infty < x < 1$

28. If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is

(a) $\left(\dfrac{1}{2}\right)^{x(x-1)}$ (b) $\dfrac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$ (c) $\dfrac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (d) not defined

29. If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval

(a) $(2, \infty)$ (b) (1, 2) (c) (-2, -1) (d) none of these

30. If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin\sqrt{x})^2$, then

(a) $f(x) = \sin^2 x,\ g(x) = \sqrt{x}$ (b) $f(x) = \sin x,\ g(x) = |x|$

(c) $f(x) = x^2,\ g(x) = \sin\sqrt{x}$ (d) f and g cannot be determined

31. Let $g(x)=1+x-[x]$ and $f(x)=\begin{cases}-1, & x<0\\ 0, & x=0\\ 1, & x>0\end{cases}$. Then for all x, f(g(x)) is equal to

(a) x   (b) 1   (c) f(x)   (d) g(x).

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32. The domain of definition of $f(x)=\dfrac{\log_2(x+3)}{x^2+3x+2}$ is

(a) R ~ {-1, -2}   (b) $(-2, \infty)$ (c) R ~ {-1, -2, -3}   (d) $(-3, \infty) \sim \{-1, -2\}$

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33. If $f(x)=\dfrac{\alpha x}{x+1}$, $x \neq -1$, then for what value of α is $f(f(x)) = x$?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$ (c) 1   (d) -1

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34. The set of all real numbers x for which $x^2 - |x+2| + x > 0$, is

(a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$

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35. Range of the function $f(x)=\dfrac{x^2+x+2}{x^2+x+1}$, $x \in R$ is

(a) $(1, \infty)$ (b) $\left(1, \dfrac{11}{7}\right)$ (c) $\left(1, \dfrac{7}{3}\right]$ (d) $\left[1, \dfrac{7}{5}\right]$

Exercise level 3 :

1. Find the domain of the following functions, where [.]denotes the greatest integer function.
(i) $f(X) = \dfrac{3}{\left[\dfrac{x}{2}\right]} - 5^{\cos^{-1} x^2} + \dfrac{(2x+1)!}{\sqrt{x+1}}$
(ii) $f(x) = \dfrac{1}{[|x-1|] + [|12-x|] - 11}$
2. (a) Find the domain and range of $f(x) = log\dfrac{1}{\sqrt{[\cos x] - [\sin x]}}$,
[.]denotes the greatest integer function.
(b). Solve the equation $||x^2 - 5x + 4| - |2x - 3|| = |x^2 - 3x + 1|$
(c). Find domain and range of function $f(x) = \sqrt{\log_{1/2} \log_2 [x^2 + 4x + 5]}$where [.]denotes the greatest integer function.
3. If a function is defined as $f(x) = \sqrt{\log_{\phi(x)} g(x)}$, where $g(x) = |\sin x| + \sin x, \phi(x) = \sin x + \cos x, 0 \leq x \leq \pi$, Then find the domain of f (x).
4(a). Find the range of the function $f(x) = \sqrt{\sin\left(\ln\left(\dfrac{x^2+e}{x^2+1}\right)\right)} + \sqrt{\cos\left(\ln\left(\dfrac{x^2+e}{x^2+1}\right)\right)}$
(b). Find the range of function $f: R \to R$ defined by $f(x) = \left(\tan^{-1} x\right)^2 + \dfrac{2}{\sqrt{1+x^2}}$
5(a). Find period of the functions
(i) $f(x) = e^{x-[x]+|\sin \pi x|+|\sin 2\pi x|+....+|\sin n\pi x|}$, [.]denotes the greatest integer.
(ii) $f(x) = \dfrac{1}{|\sin x| + |\cos x|}$
(b). Prove that $f(x) = \dfrac{2x(\sin x + \tan x)}{2\left[\dfrac{x+2\pi}{\pi}\right] - 3}$is an odd functin. [.]dentoes greatest integer function.
6. Let $f(x,y)$be a periodic function, satisfying the condition
$f(x,y) = f(2x+2y, 2y-2x)\ \forall x, y \in R$and let g (x) be a function defined as $g(x) = f(2^x, 0)$,
Prove that g (x) is periodic function and find its period,
7(a). Solve the equation $x^3 - [x] = 5$, where [.]denotes the integral part of the number x.
(b). Find all the real solutions of the equation $3x^2 - 8[x] + 1 = 0$, where [.]denotes the greatest integer function.
8. Prove that $[x] = \left[\dfrac{x}{2}\right] + \left[\dfrac{x+1}{2}\right]$where [.]denote greatest integer function and n be any positive integer then show that $\left[\dfrac{n+1}{2}\right] + \left[\dfrac{n+2}{4}\right] + \left[\dfrac{n+4}{8}\right] + \left[\dfrac{n+8}{16}\right] + ... = n$
9(a). A function f (x) is defined for all $x \in R$ and satisfies, $f(x+y) = f(x) + 2y^2 + kxy \forall x, y \in R$, where k is a given constant. If $f(1) = 2$and $f(2) = 8$,find $f(x)$and show that $f(x+y) . f\left(\dfrac{1}{x+y}\right) = k, x+y \neq 0$.
(b). Find all $f(x)$satisfying the relation $\{f(x)\}^2 + 4f(x).f'(x) + \{f'(x)\}^2 = 0$.
10. Find all possible values of real parameter 'a' so that the range of $f(x) = \dfrac{x-1}{a - x^2 + 1}$does not contain any value belonging to the interval $\left[-1, -\dfrac{1}{3}\right]$.

Exercise level 1 :

ANSWER KEY LEVEL 1

1.	b	21.	b
2.	b	22.	b
3.	b	23.	b
4.	a	24.	d
5.	a	25.	a
6.	a	26.	c
7.	c	27.	a
8.	c	28.	c
9.	b	29.	a
10.	a	30.	c
11.	b	31.	a
12.	c	32.	a
13.	c	33.	a
14.	b	34.	c
15.	d	35.	c
16.	b	36.	c
17.	b	37.	a
18.	b	38.	a
19.	d	39.	d
20.	b	40.	d

Exercise level 2:

1.	d	19.	a
2.	c	20.	a
3.	b	21.	b
4.	a	22.	a
5.	a	23.	b
6.	b	24.	b
7.	c	25.	a
8.	c	26.	c
9.	a	27.	d
10.	c	28.	b
11.	b	29.	a
12.	b	30.	a
13.	d	31.	b
14.	d	32.	d
15.	b	33.	d
16.	a	34.	b
17.	a	35.	c
18.	a		